

Sequences with e.g.f. $G(e^x - 1)$ are eventually periodic modulo m with period dividing $\varphi(m)$: a proof of Bala’s conjecture, and of the twenty OEIS instances of it

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Abstract

In July 2022 Peter Bala conjectured on OEIS A000670 that the Fubini numbers reduced modulo k are eventually periodic with period dividing $\varphi(k)$, and, more generally, that the same holds for any integer sequence whose exponential generating function has the form $G(e^x - 1)$ with G an integral power series. We prove the general statement. The proof is short: expanding $(e^x - 1)^k$ in Stirling numbers of the second kind writes $a(n) = \sum_k c_k k! S(n, k)$; modulo m every term with $k \geq m$ vanishes because $m \mid k!$, leaving a fixed finite sum; and the identity $k! S(n, k) = \sum_j (-1)^{k-j} \binom{k}{j} j^n$ turns that finite sum into an integer combination of the sequences $n \mapsto j^n$, each of which is eventually periodic modulo m with period dividing $\varphi(m)$. The same conjecture is currently posted, separately, on twenty OEIS entries; all twenty follow, and we identify the relevant G for each. We also separate off fourteen entries carrying the same wording whose generating functions are *not* of this form, where the conjecture remains open.

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1 The conjecture

Throughout, m and k denote positive integers, φ is Euler’s totient function, and $S(n, k)$ are the Stirling numbers of the second kind, with the conventions $S(0, 0) = 1$ and $S(n, k) = 0$ for $k > n$. We write $0^0 = 1$.

OEIS A000670, the Fubini numbers, carries the following comment:

Conjecture: Let k be a positive integer. The sequence obtained by reducing $a(n)$ modulo k is eventually periodic with the period dividing $\varphi(k) = A000010(k)$. For example, modulo 16 we obtain the sequence $[1, 1, 3, 13, 11, 13, 11, 13, 11, 13, \dots]$, with an apparent period of 2 beginning at $a(4)$. Cf. A354242.

More generally, we conjecture that the same property holds for integer sequences having an e.g.f. of the form $G(\exp(x) - 1)$, where $G(x)$ is an integral power series. — *Peter Bala*, Jul 08 2022

The identical general statement appears on A004123, and the specific statement appears on eighteen further entries; the full list is Table 1. As of the “Last modified” lines on the live entries (A000670 at 14 August 2026, revision 909; A004123 at 30 May 2026, revision 98) the statements are still recorded as conjectures, and no proof appears among the entries’ links, formulas or programs.

A related note of Bala, *Integer sequences that become periodic on reduction modulo k for all k* (December 2017, linked from A047974), proves a different theorem: there the generating function has the shape $F(x) \exp(xG(x))$ and the conclusion is $a(n+k) \equiv a(n) \pmod{k}$, i.e. pure periodicity with period dividing k . Neither the hypothesis nor the conclusion of that theorem covers the present conjecture, and the argument below is unrelated to it.

We prove:

Theorem 1. *Let $G(y) = \sum_{k \geq 0} c_k y^k$ be a power series with integer coefficients, and define integers $a(n)$ by*

$$\sum_{n \geq 0} a(n) \frac{x^n}{n!} = G(e^x - 1).$$

Then for every $m \geq 1$ the sequence $(a(n) \bmod m)_{n \geq 0}$ is eventually periodic with period dividing $\varphi(m)$.

2 Three standard ingredients

Lemma 2. *For every $k \geq 0$, $(e^x - 1)^k = k! \sum_{n \geq k} S(n, k) \frac{x^n}{n!}$, and consequently*

$$a(n) = \sum_{k=0}^n c_k k! S(n, k) \quad (n \geq 0).$$

In particular each $a(n)$ is an integer, the sum being finite because $S(n, k) = 0$ for $k > n$.

Proof. The first identity is the standard exponential generating function of the Stirling numbers of the second kind. Multiplying by c_k and summing over k gives $G(e^x - 1) = \sum_k c_k k! \sum_n S(n, k) x^n / n!$; the interchange is legitimate in $\mathbb{Z}[[x]]$ because $(e^x - 1)^k$ has order k in x , so only finitely many k contribute to each coefficient. Comparing coefficients of $x^n / n!$ gives the formula. $\square$

Lemma 3. *For all $n, k \geq 0$, $k! S(n, k) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$.*

Proof. This is inclusion–exclusion for surjections: $k! S(n, k)$ counts the surjections from an n -set onto a k -set, and the right-hand side counts them by excluding the images that miss at least one point. (For $n = 0$ both sides are $[k = 0]$, with $0^0 = 1$.) $\square$

Lemma 4. *Let $m \geq 1$ and $j \geq 0$ be integers. Then $(j^n \bmod m)_{n \geq 0}$ is eventually periodic with period dividing $\varphi(m)$.*

Proof. Factor $m = m_1 m_2$, where $m_1 = \prod_{p | \gcd(j, m)} p^{v_p(m)}$ collects the prime powers of m whose prime divides j , and $m_2 = m / m_1$. Then $\gcd(m_1, m_2) = 1$ and $\gcd(j, m_2) = 1$.

Modulo m_1 : every prime dividing m_1 divides j , so if v denotes the largest exponent occurring in m_1 then $m_1 \mid j^n$ for all $n \geq v \cdot \omega(m_1)$; more simply, $m_1 \mid j^n$ for all $n \geq \log_2 m$. Hence $j^n \equiv 0 \pmod{m_1}$ for all large n : the sequence is eventually constant, of eventual period 1.

Modulo m_2 : j is a unit, so j^n is purely periodic with period $\text{ord}_{m_2}(j)$, which divides $\varphi(m_2)$.

By the Chinese remainder theorem, $j^n \bmod m$ is eventually periodic with period dividing $\text{lcm}(1, \text{ord}_{m_2}(j)) = \text{ord}_{m_2}(j)$. Since $m_2 \mid m$ we have $\varphi(m_2) \mid \varphi(m)$, so this period divides $\varphi(m)$. $\square$

3 Proof of Theorem 1

Proof. Fix $m \geq 1$. Since $m \mid m!$, we have $m \mid k!$ for every $k \geq m$, so every term of Lemma 2 with $k \geq m$ vanishes modulo m . Hence for every $n \geq m$

$$a(n) \equiv \sum_{k=0}^{m-1} c_k k! S(n, k) \pmod{m}, \quad (1)$$

and the right-hand side is a sum with a *fixed* number of terms, independent of n . This is the one place where the hypothesis $G \in \mathbb{Z}[[y]]$ is used: it guarantees that the coefficients c_k are integers, so that the factor $k!$ — and with it the divisibility by m — is not disturbed by a denominator.

Substituting Lemma 3 into (1) and exchanging the two finite sums,

$$a(n) \equiv \sum_{k=0}^{m-1} c_k \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = \sum_{j=0}^{m-1} w_j j^n \pmod{m}, \quad w_j := \sum_{k=j}^{m-1} (-1)^{k-j} \binom{k}{j} c_k \in \mathbb{Z}, \quad (2)$$

for all $n \geq m$. The weights w_j depend on m and on G , but not on n .

By Lemma 4 each of the m sequences $n \mapsto j^n \pmod{m}$, for $0 \leq j \leq m-1$, is eventually periodic with period dividing $\varphi(m)$. A finite $\mathbb{Z}$ -linear combination of sequences that are eventually periodic with periods $p_0, \dots, p_{m-1}$ is eventually periodic with period dividing $\text{lcm}(p_0, \dots, p_{m-1})$; here each p_j divides $\varphi(m)$, and the least common multiple of finitely many divisors of $\varphi(m)$ again divides $\varphi(m)$. Hence the right-hand side of (2) is eventually periodic modulo m with period dividing $\varphi(m)$, and by (2) so is $a(n)$. $\square$

Remark 5. “Eventually” cannot be dropped. For the Fubini numbers modulo 16 the sequence begins 1, 1, 3, 13, 11, 13, 11, ... and is periodic only from $n = 4$ onwards, exactly as Bala’s comment records; the initial segment 1, 1, 3 is not part of any period. The proof shows where this comes from: (1) is asserted only for $n \geq m$, and Lemma 4 contributes a further preperiod from the primes shared by j and m .

Remark 6. The bound $\varphi(m)$ is not always attained, and the theorem does not claim it is. For the Fubini numbers modulo 16 the exact period is 2 while $\varphi(16) = 8$; modulo 25 the exact period is 20 = $\varphi(25)$. Both are consistent with, and predicted by, (2): the exact period is the least common multiple of the orders $\text{ord}_{m_2}(j)$ over those j whose weight w_j survives modulo m .

4 The twenty entries

Theorem 1 applies to an entry as soon as its generating function is exhibited in the form $G(e^x - 1)$ with $G \in \mathbb{Z}[[y]]$. Table 1 does this for every OEIS entry we found carrying the conjecture in the $\varphi(k)$ form. In each row, y abbreviates $e^x - 1$, so that $e^x = 1 + y$; the reduction is then elementary algebra, and the resulting G is visibly integral.

Table 1: The twenty entries settled by Theorem 1.

entry	e.g.f. as posted	$G(y)$, $y = e^x - 1$	$c_0, c_1, c_2, \dots$
A000670	$1/(2 - e^x)$	$1/(1 - y)$	1, 1, 1, 1, ...
A002050	$(e^{2x} - e^x)/(2 - e^x)$	$y(1 + y)/(1 - y)$	0, 1, 2, 2, 2, ...
A004123	$1/(3 - 2e^x)$	$1/(1 - 2y)$	1, 2, 4, 8, ...
A006531	$C(1 - e^{-x})$, C the Catalan o.g.f.	$C(y/(1 + y))$	1, 1, 1, 2, 4, 9
A052895	$C(e^x - 1)$, C the Catalan o.g.f.	$C(y)$	1, 1, 2, 5, 14, 42
A064618	Stirling transform of $(n!)^2$	$\sum_k k! y^k$	1, 1, 2, 6, 24, 120

entry	e.g.f. as posted	$G(y), y = e^x - 1$	$c_0, c_1, c_2, \dots$
A080253	$e^x/(2 - e^{2x})$	$(1 + y)/(1 - 2y - y^2)$	1, 3, 7, 17, 41, 99
A162314	$e^{2x}/(2 - e^{2x})$	$(1 + y)^2/(1 - 2y - y^2)$	1, 4, 10, 24, 58, 140
A167137	$P(e^x - 1)$, P the partition g.f.	$P(y)$	1, 1, 2, 3, 5, 7
A259533	$e^{3x}/(2 - e^x)$	$(1 + y)^3/(1 - y)$	1, 4, 7, 8, 8, 8
A301921	$1/(1 - (e^x - 1)/(1 - (e^x - 1)^2/\dots))$	continued fraction in y	1, 1, 1, 2, 3, 5
A305550	$\prod_{k \geq 1} (1 + (e^x - 1)^k)$	$\prod_{k \geq 1} (1 + y^k)$	1, 1, 1, 2, 2, 3
A306082	$\prod_{k \geq 1} 1/(1 - (e^x - 1)^{k^2})$	$\prod_{k \geq 1} 1/(1 - y^{k^2})$	1, 1, 1, 1, 2, 2
A316142	$\prod_{k \geq 1} (1 + (e^x - 1)^{k^2})$	$\prod_{k \geq 1} (1 + y^{k^2})$	1, 2, 3, 6, 9, 14
A316143	$\prod_{k \geq 1} 1/(1 - (e^x - 1)^{k^2})$	$\prod_{k \geq 1} 1/(1 - y^{k^2})$	1, 2, 5, 10, 20, 36
A316144	$\prod_{k \geq 1} ((1 + (e^x - 1)^k)/(1 - (e^x - 1)^k))$	$\prod_{k \geq 1} (1 + y^k)/(1 - y^k)$	1, 4, 12, 32, 76, 168
A320352	$(e^x - 1)/(e^x - e^{2x} + 1)$	$y/(1 - y - y^2)$	0, 1, 1, 2, 3, 5
A354242	$1/\sqrt{5 - 4e^x}$	$(1 - 4y)^{-1/2}$	1, 2, 6, 20, 70, 252
A354253	$1/\sqrt{9 - 8e^x}$	$(1 - 8y)^{-1/2}$	1, 4, 24, 160, 1120, 8064
A355409	$1/(1 + e^{2x} - e^{3x})$	$1/(1 - y - 2y^2 - y^3)$	1, 1, 3, 6, 13, 28

Three rows deserve a word. For A354242 and A354253 the substitution gives $(1 - 4y)^{-1/2}$ and $(1 - 8y)^{-1/2}$, whose coefficients are $\binom{2k}{k}$ and $2^k \binom{2k}{k}$; these are integers, so G is integral even though the posted expression involves a square root. For A006531 the posted e.g.f. is $C(1 - e^{-x})$ rather than a function of $e^x - 1$; but $1 - e^{-x} = y/(1 + y)$, and $C(y/(1 + y))$ is an integral power series in y , so the hypothesis holds after all. A004123 has offset 1 and its own e.g.f. line specifies “e.g.f. for sequence with offset 0”, i.e. the entry’s $a(n)$ is the shift of the sequence generated by $1/(1 - 2y)$; a shift changes neither eventual periodicity nor the period.

Corollary 7. *The conjecture as posted is true on each of the twenty entries of Table 1, and the general form stated on A000670 and A004123 is true as stated.*

5 Fourteen entries where this argument does not apply

The same conjecture wording appears on further entries, and Theorem 1 says nothing about them. We list them so that they are not mistakenly treated as settled: A000657, A002105, A012780 ($\arcsin(\tan x)$), A126156 ($\sqrt{\sec(\sqrt{2}x)}$), A143138 and A143139 (implicitly defined e.g.f.s), together with A258899, A258903, A276371, A365777, A365782, A365794 and A370092, whose posted e.g.f.s involve $\exp(x^n)$ or fractional powers.

For each of these we recovered the putative coefficients c_k from the published terms by inverting the substitution: if $A(x) = G(e^x - 1)$ then $x = \log(1 + y)$ gives

$$c_k = \frac{1}{k!} \sum_n s(k, n) a(n), \quad (3)$$

with s the signed Stirling numbers of the first kind. In every case some c_k failed to be an integer, at the value of k recorded in Section 6. So these thirteen entries genuinely lie outside the family, and their conjectures remain open.

One further entry we leave explicitly undecided, and flag rather than quietly count. A000436 (generalized Euler numbers $c(3, n)$) has posted e.g.f. $\cos x / \cos 3x$ taken in even powers only, which is not of the form $G(e^x - 1)$. But the sequence listed on the entry is the reindexed even part, whose own exponential generating function we have no closed form for, and the coefficients recovered from (3) are integers for every k we could test — all thirteen published terms, and all twenty-two coefficients obtainable after extending the terms from $\cos x / \cos 3x$ by exact series arithmetic. So A000436 may well satisfy the hypothesis of Theorem 1. We have not proved that it does, and we do not claim its conjecture.

6 Verification

All numerical claims were produced by a script written separately from this note, and every number quoted above is one of its outputs.

For each entry, the coefficients c_k were recovered from the published DATA by the inversion $c_k = \frac{1}{k!} \sum_n s(k, n) a(n)$ and tested for integrality. The twenty entries of Table 1 passed over the full published range; the values shown in the last column are the recovered c_k , and they agree with the G read off in the third column in every case. The thirteen entries of the previous section failed, first at $k = 2$ (A000657, A002105), $k = 3$ (A258899, A370092), $k = 4$ (A258903, A276371, A365777, A365782, A365794, A012780, A126156, A143139) and $k = 6$ (A143138). A000436 did not fail: its recovered c_k were integral for all thirteen published terms, and for all twenty-two coefficients obtainable after extending the terms from $\cos x / \cos 3x$ by exact series arithmetic. That is why it is left undecided above rather than excluded.

The conjecture itself was then checked directly. For each of the twenty entries we generated $a(n) \bmod m$ for $n \leq 400$ from the recovered c_k using $a(n) = \sum_k c_k k! S(n, k)$ with an integer recurrence for $S(n, k)$ — a route that touches neither the e.g.f. nor the published terms — measured the exact eventual period over the last 160 values, and confirmed that it divides $\varphi(m)$, for $m \in \{5, 7, 8, 9, 11, 12, 13, 16, 25, 27\}$. It does so in all 200 entry-modulus pairs.

As an independent check of the generation itself, the Fubini case was also computed by exact rational power-series arithmetic from $G(e^x - 1)$ directly, for $n \leq 40$, and reduced modulo each m ; it agrees with the Stirling-form computation throughout. The two worked examples quoted in Section 1 were reproduced: modulo 16 the Fubini numbers give 1, 1, 3, 13, 11, 13, 11, 13, $\dots$, exact period 2 from $n = 4$, and $\varphi(16) = 8$; modulo 25 the exact period is $20 = \varphi(25)$.

7 Concluding remarks

The mechanism is worth isolating, because it explains why φ appears at all in a statement about exponential generating functions. Reducing modulo m collapses the infinite Stirling expansion to its first m terms, because $m \mid k!$ for $k \geq m$; and the first m terms are, by inclusion–exclusion, an integer combination of the pure exponentials j^n with $j < m$. Once the sequence is written that way it is governed entirely by multiplicative orders modulo m , and every such order divides $\varphi(m)$. The hypothesis that G be integral is exactly what licenses the first step.

The same argument gives a slightly stronger statement than the one conjectured: the eventual period divides $\text{lcm}\{\text{ord}_{m_2}(j)\}$ over the finitely many $j < m$ with $w_j \not\equiv 0$, and the preperiod is at most m . It also shows the result is insensitive to shifting or to adding a polynomial-in- n correction, since those operations preserve the exponential form (2).

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequences A000670, A002050, A004123, A006531, A052895, A064618, A080253, A162314, A167137, A259533, A301921, A305550, A306082, A316142, A316143, A316144, A320352, A354242, A354253, A355409; conjectures of P. Bala, 2017–2023.
- [2] P. Bala, *Integer sequences that become periodic on reduction modulo k for all k* , December 2017; linked from OEIS A047974.
- [3] L. Comtet, *Advanced Combinatorics*, Reidel, 1974, Chapter V (Stirling numbers, surjections, and the expansion of $(e^x - 1)^k$).